

TECHNICAL PRESENTATION

AI-POWERED SMART RETAIL STORE USING COMPUTER VISION

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Course: Computer Vision(ITA 0509)

INTRODUCTION

What is Computer Vision?

- Computer Vision is a field of Artificial Intelligence (AI).
- It enables computers to understand and analyze images and videos.
- In retail, cameras and AI models can analyze:
 - Customers
 - Products
 - Shelves
 - Store activities
 - Shopping behaviour

Objective:

- To develop an intelligent retail system that uses Computer Vision to improve customer experience, inventory management, and store security.



PROBLEM STATEMENT

Existing Retail Challenges

Traditional retail stores face several problems:

- Manual inventory monitoring
- Products going out of stock
- Long checkout queues
- Difficulty in monitoring customer behaviour
- Manual shelf inspection
- Product misplacement
- Theft and suspicious activities
- Lack of real-time store analytics



RETAIL APPLICATIONS OF COMPUTER VISION

1. Product Detection

Identifies products using object detection.

2. Smart Shelf Monitoring

Detects:

Empty shelves

Low stock

Wrong product placement

3. Automated Checkout

Tracks selected products and reduces checkout time.

4. Customer Behaviour Analysis

Analyses:

Customer movement

Store traffic

Popular product areas

5. Security & Loss Prevention

Detects unusual or suspicious activities.

6. Queue Monitoring

Detects long queues and alerts staff.

SYSTEM ARCHITECTURE

INPUT



IMAGE PROCESSING



COMPUTER VISION



ANALYSIS



OUTPUT



CONCLUSION

- **Smart Vision Retail** demonstrates how Computer Vision can transform traditional retail stores into intelligent and automated environments.
- By combining **Computer Vision, Deep Learning, Object Detection, and Real-Time Analytics**, the system can improve:
 - Store efficiency
 - Inventory monitoring
 - Security
 - Customer experience
 - Decision making

"Computer Vision enables retail stores to see, understand, and respond intelligently."

*Thank
you!*